

Linear algebra and geometry 1 - TTPU

Started on Wednesday, 27 January 2021, 6:42 AM

State Finished

Completed on Wednesday, 27 January 2021, 6:43 AM

Time taken 1 min 22 secs

Marks 80.00/1000.00

Grade 2.64 out of 33.00 (8%)

Question 1

Correct

Mark 100.00 out of 100.00

Fixed a coordinate system in the space, consider the points of coordinates

$P_0 = (1, 1, 1)$, $P_1 = (2, 0, 2)$ e $P_2 = (0, 0, 0)$. Then, the area of the triangle of vertices P_0 , P_1 and P_2 is equal to

- ☐ (a) 2.
- ☒ (b) $\sqrt{2}$. ✓
- ☐ (c) $\frac{1}{\sqrt{2}}$.
- ☐ (d) $2\sqrt{2}$.

Risposta corretta.

The correct answer is: $\sqrt{2}$.

Question 2

Correct

Mark 100.00 out of
100.00

Consider the following line $r : 2x - y - z = 2x + y + 2z - 1 = 0$.

Find the correct statement.

-
- ☐ (a) r passes through the origin.
- ☒ (b) r is parallel to the line $s(t) = (10 - t, 1 - 6t, 4t)$. ✓
- ☐ (c) r lies on the plane $4x + z - 2 = 0$.
- ☐ (d) r lies on the plane yz .

Risposta corretta.

The correct answer is: r is parallel to the line $s(t) = (10 - t, 1 - 6t, 4t)$.

Question 3

Incorrect

Mark -15.00 out of
100.00

Consider the following linear system in the variables x, y, z :

$$\begin{cases} x - 2y + z = 2 \\ -2x + y = 4 \\ x - z = 1 \end{cases}$$

Find the right statement.

-
- ☐ (a) The system has only two solutions.
- ☒ (b) The system has infinitely many solutions. ✗
- ☐ (c) The system has one unique solution.
- ☐ (d) The system has no solutions.

Risposta errata.

The correct answer is: The system has one unique solution.

Question 4

Incorrect

Mark -15.00 out of
100.00

Let us take the matrix $A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \end{pmatrix}$; find the right statement:

- ☒ (a) $A^{-1} = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{pmatrix}$. ✖
- ☐ (b) $A^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{pmatrix}$.
- ☐ (c) $A^{-1} = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \end{pmatrix}$.
- ☐ (d) $A^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$.

Risposta errata.

The correct answer is: $A^{-1} = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \end{pmatrix}$.

Question 5

Incorrect

Mark -15.00 out of
100.00

Consider a linear mapping $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$. Find which one of the following statements cannot hold.

- ☒ (a) $\dim \ker(f) = 0$. ✖
- ☐ (b) The matrix associated to f has rank 1.
- ☐ (c) f is surjective.
- ☐ (d) $\dim \ker(f) \geq 1$.

Risposta errata.

The correct answer is: f is surjective.

Question 6

Incorrect

Mark -15.00 out of
100.00

Let us consider the vectors $\vec{v}_1 = (1, 1, 1, 0)$, $\vec{v}_2 = (2, 2, 0, 2)$, $\vec{v}_3 = (3, 0, 3, 3)$ e $\vec{v}_4 = (0, 4, 4, 4)$ in $\mathbb{R}^4$. Find the right statement.

- ☒ (a) There exist $a, b, c \in \mathbb{R}$ such that $\vec{v}_4 = a\vec{v}_1 + b\vec{v}_2 + c\vec{v}_3$. ✖
- ☐ (b) $\{\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4\}$ is a basis of $\mathbb{R}^4$.
- ☐ (c) The vector $\vec{v}_1$ lies in the subspace generated by $\{\vec{v}_2, \vec{v}_3, \vec{v}_4\}$.
- ☐ (d) $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is a basis of $\mathbb{R}^2$.

Risposta errata.

The correct answer is: $\{\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4\}$ is a basis of $\mathbb{R}^4$.**Question 7**

Incorrect

Mark -15.00 out of
100.00

Let $q(x, y) = -x^2 - \sqrt{2}xy + y^2$ be a quadratic form. Find the correct statement.

- ☐ (a) q is positive definite.
- ☐ (b) q is indefinite.
- ☒ (c) The matrix associated to q is $\begin{pmatrix} 1 & -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -1 \end{pmatrix}$. ✖
- ☐ (d) q is negative definite.

Risposta errata.

The correct answer is: q is indefinite.

Question 8

Incorrect

Mark -15.00 out of 100.00

Let us consider the conic $\mathcal{C}$ in the Euclidean plane defined by the equation $2x^2 + 4xy + 2x + 2y^2 + 3 = 0$.

Find the correct statement.

-
- ☒ (a) $\mathcal{C}$ is degenerate. ✖
- ☐ (b) None of the three other statements is correct.
- ☐ (c) $\mathcal{C}$ is a parabola.
- ☐ (d) $\mathcal{C}$ is an ellipse.

Risposta errata.

The correct answer is: $\mathcal{C}$ is a parabola.

Question 9

Incorrect

Mark -15.00 out of 100.00

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $f(x, y, z) = (x + 3y + 4z, x + 3y + 4z, x + 3y + 4z)$. Let λ_i be an eigenvalue of f and let $E(\lambda_i)$ be the corresponding eigenspace.

Find the correct statement.

-
- ☒ (a) $\lambda_1 = 3, \lambda_2 = 1, \lambda_3 = 0, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 2, \dim E(\lambda_3) = 1$ ✖
- ☐ (b) $\lambda_1 = 3, \lambda_2 = 1, \lambda_3 = 7, \dim E(\lambda_1) = 2, \dim E(\lambda_2) = 1, \dim E(\lambda_3) = 1$
- ☐ (c) $\lambda_1 = 8, \lambda_2 = 0, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 2$
- ☐ (d) $\lambda_1 = 3, \lambda_2 = 4, \lambda_3 = 1, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 1, \dim E(\lambda_3) = 1$
- ☐ (e) $\lambda_1 = 4, \lambda_2 = 0, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 2$
- ☐ (f) $\lambda_1 = 8, \lambda_2 = 1, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 2$

Risposta errata.

The correct answer is: $\lambda_1 = 8, \lambda_2 = 0, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 2$

Question 10

Incorrect

Mark -15.00 out of
100.00

For $a, b \in \mathbb{R}$ consider the following linear system in the variables x, y, z :

$$\begin{cases} x - by + z = 0 \\ x + z = 0 \\ ax - y + 2z = 0 \end{cases}$$

and let $S_{a,b}$ be the set of solutions of the linear system of equations.

Find the right statement.

-
- ☒ (a) For $a = 2$ and $b = 3$ the linear system admits infinitely many solutions and $\dim S_{a,b} = 3$. ✖
 - ☐ (b) For $a = 2$ and $b = 3$ the linear system does not have solutions.
 - ☐ (c) For $a = 2$ and $b = 1$ the linear system admits infinitely many solutions and $\dim S_{a,b} = 2$.
 - ☐ (d) For $a = 1$ and $b = 3$ the linear system admits infinitely many solutions and $\dim S_{a,b} = 1$.
 - ☐ (e) For $a = 2$ and $b = 1$ the linear system admits infinitely many solutions and $\dim S_{a,b} = 1$.
 - ☐ (f) For $a = 1$ and $b = 3$ the linear system admits infinitely many solutions and $\dim S_{a,b} = 2$.

Risposta errata.

The correct answer is: For $a = 2$ and $b = 1$ the linear system admits infinitely many solutions and $\dim S_{a,b} = 1$.